

EXPERIMENT 11

Student Name: Punya Arora

Branch: AIT-CSE-AIML

Semester: 5

Subject Name: ADBMS

UID: 24BAI70928

Section/Group: 24AIT_KRG1_B

Subject Code: 24CSP-310

Question 1:

Question:

A company stores the monthly salary history of its employees in an EmployeeSalaryHistory table.

Write an SQL query to find the employees whose salary has increased in every recorded month compared to the immediately previous month. For each salary record, compare the current month's salary with the employee's previous month's salary using a Window Function. Display only those employees for whom all later salary records are greater than their respective previous salary.

Output should display:

- Employee ID
- Employee Name

Answer:

```
SELECT DISTINCT employee_id, employee_name
```

```
FROM (
```

```
    SELECT
```

```
        employee_id,
```

```
        employee_name,
```

```
        salary,
```

```
        LAG(salary) OVER (
```

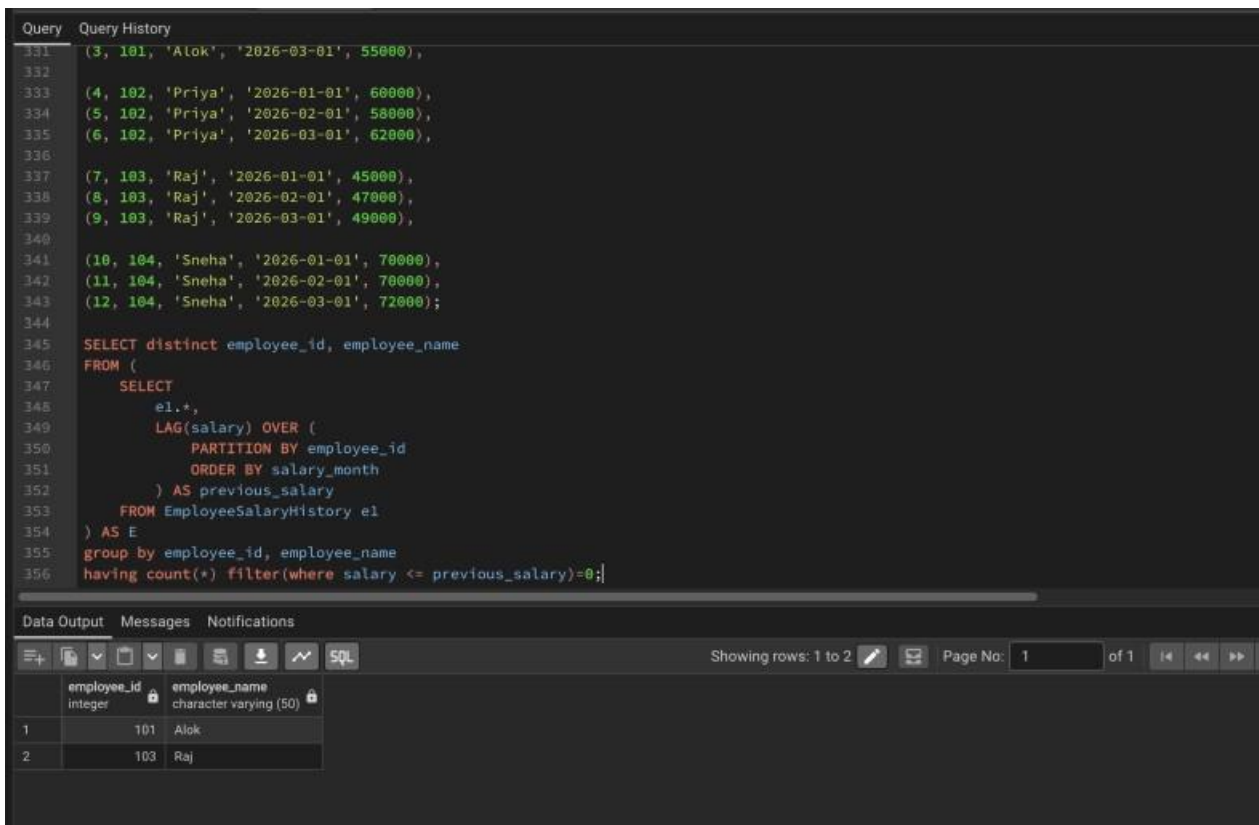
```
            PARTITION BY employee_id
```

EXPERIMENT 11

```

ORDER BY salary_month
) AS previous_salary
FROM EmployeeSalaryHistory
) AS E
GROUP BY employee_id, employee_name
HAVING COUNT(*) FILTER (
WHERE salary <= previous_salary
) = 0;

```



The screenshot shows a SQL IDE with a query editor and a data output window. The query editor contains a complex SQL query using window functions to find employees whose current salary is less than or equal to their previous salary. The data output window shows the results of the query, which are two rows of employee data.

```

Query History
331 (3, 101, 'Alok', '2026-03-01', 55000),
332
333 (4, 102, 'Priya', '2026-01-01', 60000),
334 (5, 102, 'Priya', '2026-02-01', 58000),
335 (6, 102, 'Priya', '2026-03-01', 62000),
336
337 (7, 103, 'Raj', '2026-01-01', 45000),
338 (8, 103, 'Raj', '2026-02-01', 47000),
339 (9, 103, 'Raj', '2026-03-01', 49000),
340
341 (10, 104, 'Sneha', '2026-01-01', 70000),
342 (11, 104, 'Sneha', '2026-02-01', 70000),
343 (12, 104, 'Sneha', '2026-03-01', 72000);
344
345 SELECT distinct employee_id, employee_name
346 FROM (
347     SELECT
348         e1.*,
349         LAG(salary) OVER (
350             PARTITION BY employee_id
351             ORDER BY salary_month
352         ) AS previous_salary
353     FROM EmployeeSalaryHistory e1
354 ) AS E
355 group by employee_id, employee_name
356 having count(*) filter(where salary <= previous_salary)=0;

```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

employee_id	employee_name
101	Alok
103	Raj

Question 2:

Question:

EXPERIMENT 11

A company maintains employee details in an Employee table and department information in a Department table.

Write an SQL query to find the employees whose salaries belong to the **top three distinct salary levels** within their respective departments.

If multiple employees have the same salary, they should receive the same salary rank and all of them should be included in the result.

Output should display:

- Department Name
- Employee Name
- Salary **Answer:**

SELECT

D.name AS Department,

E.name AS Employee,

E.salary

FROM (

SELECT

*,

DENSE_RANK() OVER (

PARTITION BY departmentId

ORDER BY salary DESC

) AS rnk

FROM Employee)

AS E

JOIN Department D

ON E.departmentId = D.id

EXPERIMENT 11

WHERE $rnk \leq 3$;

```

397
398 --Solution:
399 SELECT D.name AS Department,
400        E.name AS Employee,
401        E.salary
402 FROM (
403     SELECT *,
404            DENSE_RANK() OVER (
405                PARTITION BY departmentId
406                ORDER BY salary DESC
407            ) AS rnk
408     FROM Employee
409 ) E
410 JOIN Department D
411 ON E.departmentId = D.id
412 WHERE rnk <= 3;
413
414

```

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1 of 1

	department character varying (50)	employee character varying (50)	salary integer
1	IT	Max	90000
2	IT	Jim	85000
3	IT	John	85000
4	IT	Alex	75000
5	Sales	Henry	80000
6	Sales	Susan	75000
7	Sales	David	70000
8	HR	Tina	60000